

StarFix

Navigate without GPS: find your direction from a photo of the stars, like a spacecraft does.

The Idea

Spacecraft use star trackers: cameras that recognize star patterns to work out which way they are pointing. Sailors used the stars to find their position. StarFix brings that to a phone. Take or upload a photo of the night sky; StarFix identifies the stars, works out where the camera was pointing, and, combined with the time and the phone's tilt, estimates where you are on Earth. Grok Voice walks you through the result.

Track Fit

Built with Cursor	Image pipeline, star matching, and UI built in Cursor.
Grok Voice / Imagine	Voice explains the fix; Imagine can produce a labeled sky chart.
Real space data	Hipparcos or Yale Bright Star catalog; real star-tracker methods.
Navigation theme	Navigation when GPS is unavailable: on Mars, in deep space, or when GPS is jammed.
Also eligible	Big Red, Software; Hardware if using a Raspberry Pi camera.

Features

MVP

- Upload a night-sky photo.
- Detect stars and match them to a catalog (“plate solving”).
- Show the photo with stars and constellations labeled, plus the pointing direction.
- Grok Voice explains: *“You were facing northeast, about 40 degrees up.”*

Stretch

- Estimate latitude and longitude from the star fix, time, and phone tilt.
- Live camera mode.
- Raspberry Pi camera as a DIY star tracker (Hardware track).

Tech Stack and Data

- Star matching: astrometry.net (local install or its web API), or Python `tetra3`.
- Image processing: Python with OpenCV / Astropy.
- Frontend: React upload page with an annotated image.
- AI: Grok Voice API (and optionally Imagine).

Limitations and Risks

- **Hardest technically.** Reliable star matching is a serious algorithm. Writing it from scratch in 24 hours is very risky.
Mitigation: Use an existing solver (astrometry.net, tetra3) and focus on the experience. Prototype this step first, Friday night.
- **Phone photos are hard.** Phones capture few stars, with noise, blur, and light pollution, especially in Ithaca in October (often cloudy).
Mitigation: Use long-exposure or night-mode photos and keep a set of good sample images ready.
- **Daytime indoor judging.** You cannot take a sky photo during judging.
Mitigation: Demo with pre-captured photos; show a short video of capturing one.
- **Location needs more than a photo.** A sky photo alone gives pointing direction, not position. Position also needs an accurate time and the camera's angle to the horizon.
Mitigation: Keep location as a stretch goal; the MVP is “which way am I pointing.”
- **Solver setup.** astrometry.net's local install needs large index files and is awkward on Windows; its web API can be slow (minutes per image).
Mitigation: Test both Friday night; pre-solve demo images and cache the results.
- **Grok fit is thinner.** Voice explains results but isn't essential to the core function.
Mitigation: Make the voice explanation educational: how sailors and spacecraft do this.

Scorecard

Demo-able	3/5	Pre-captured images only.
Buildable in 24h	2/5	Depends heavily on the solver working.
Navigation theme	5/5	Navigation without GPS.
Wow factor	5/5	“Our phone found where it was from the stars.”
Technical risk (5 = riskiest)	5/5	High.

Verdict

High risk, high reward. The most “legendary” concept, but only pick it if a quick test Friday night shows the star solver working on a phone photo.